



MINING DEALER BEST PRACTICE

Applying Heat Tabs on 3400/3500/C-Series Engine Cylinder Heads

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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1.0 Introduction

The overheating of machines in the mining industry happens on a regular basis. With the updates in technology, dealerships have been able to take advantage of that to further business with the use of ECM's (electronic control module). ECM's allow customer/technicians to check the engine/machine for any damage, abuse, available updates or reoccurring events.

Several engine types do not have an ECM (3400 & 3500) which in turn makes it very difficult from a dealer's standpoint to know if the machine is being overheated or abused. The engine heat tabs are used to track each cylinder head for overheating. The placement of the heat tabs varies depending on the style/series of engine, but primarily they are placed by the water port/thermostat housing of the cylinder head. For engines with ECM's (C-Series) using a heat tab is another way, along with the ECM, to track the overheating of the cylinder heads.

In this Best Practice we will discuss the placement and installation of engine heat tabs on 3400, 3500 and C-Series engines to better document overheating/abuse.



2.0 Best Practice Description

Many dealers that are associated with the mining industry have debated with customers about overheating and/or abuse of machines. Engine heat tabs allow for the dealer to have reinforcement due to the fact that if the heat tab is melted or missing, that specific cylinder head reached an overheating temperature in excess of 235 degrees Fahrenheit.

This Best Practice will guide dealers to show how to clean the cylinder head, prep the tab, dispense the adhesive and place the tab on the cylinder head. All of these steps will allow for a more efficient and safe work environment to reduce overheating/abuse to the engine/machine.

3.0 Implementation Steps

The placement of heat tabs vary according to which style engine is being worked on. Prior to installation the area in which the tab is to be placed needs to be cleaned thoroughly with Loctite 7387 activator. Next apply the Loctite 330 adhesive to back of heat tab. Lastly place heat tab on to designated area and hold for 30-60 seconds until solidly in place.

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C-Series – Tabs are placed on both RH side top of cylinder head and LH side top of rear (opposite) cylinder head.



3408 (MU1) - Tab is placed on the outside of the cylinder by the water jacket.



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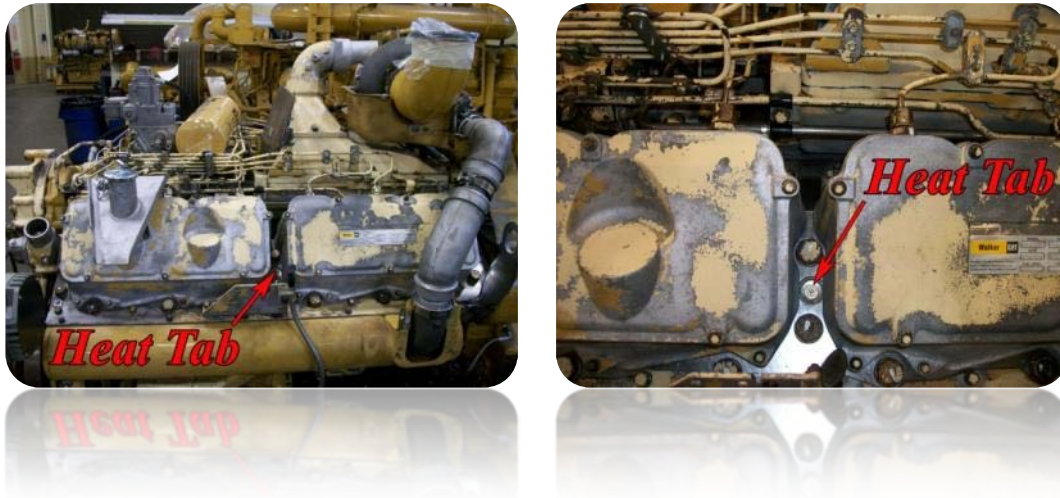
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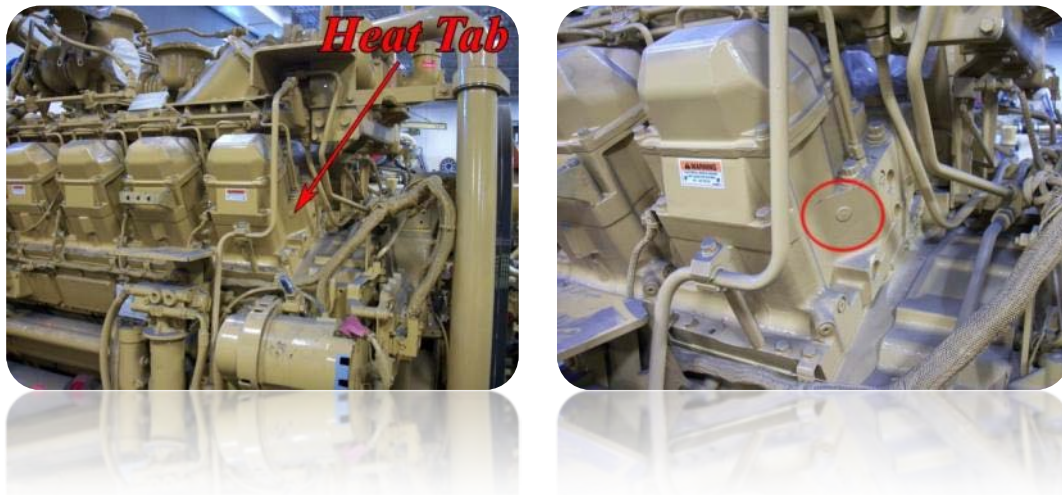
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3412 (MUI) - Tab is placed on top in middle freeze plug.



3500 – Tab placed on outside of each cylinder on flat casting surface.



4.0 Benefits

- Allows customer/dealer visual indications if cylinder heads have been overheated.

5.0 Resources Required

- Lo Temp Heat tabs
 - o Supplier – Irontite
 - o Part # 411-5811-90
- Loctite adhesive kit
 - o Loctite Adhesive Kit 330

6.0 Supporting Attachments / References

- None

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7.0 Related Best Practices

- None

8.0 Acknowledgements

- Daniel Peterson – Continuous Improvement
 - o Walker Machinery
 - o dpeterson@walker-cat.com
- Tom Williamson – Technician
 - o Walker Machinery
- Steve Winebrenner – AFA
 - o Walker Machinery

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